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CARD EDGE CONNECTOR MOUNTED ON A CIRCUIT BOARD

Abstract

A card edge connector mounted on a circuit board includes an insulating housing having two side walls extending along a longitudinal direction, plural terminals retained in the insulating housing, at least one nut fixed in the insulating housing adjacent to the side walls, and a bolt matching the nut to bolt the card edge connector to the circuit board.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to a card edge connector, and particularly to a card edge connector mounted on a circuit board.

Description of Related Arts

[0002] China Patent No. 217444661 discloses a card edge connector for mounting on a circuit board and for mating with a mating member, including an insulating body, a plurality of terminals retained in the insulating housing, a locking member mounted to the insulating body, and a shell covering the insulating housing. The locking member is used to lock and release the mating member. When in use, the locking member may cause the card edge connector to shake inevitably.

[0003] Therefore, it is desired to provide a new card edge connector.

SUMMARY OF THE INVENTION

[0004] An object of the present invention is to provide a card edge connector that is kept on the circuit board stably.

[0005] To achieve the above-mentioned object, a card edge connector mounted on a circuit board comprises an insulating housing extending along a longitudinal direction, a plurality of terminals retained in the insulating housing, at least one nut fixed in the insulating housing, and a bolt matching the nut to bolt the card edge connector to the circuit board.

[0006] Other advantages and novel features of the invention will become more apparent from the following detailed description of the present embodiment when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWING

[0007] FIG. 1 is a perspective view of a card edge connector mounted on the circuit board of an embodiment of this present invention;

[0008] FIG. 2 is another perspective view of a part of the card edge connector mounted on the circuit board in FIG. 1;

[0009] FIG. 3 is an exploded perspective view of the card edge connector mounted on the circuit board in FIG. 1;

[0010] FIG. 4 is a further exploded perspective view of the card edge connector mounted on the circuit board in FIG. 3;

[0011] FIG. 5 is a cross-sectional view of the card edge connector mounted on the circuit board taken along line A-A in FIG. 1; and

[0012] FIG. 6 is another perspective view of a card edge connector in FIG. 1;

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0013] FIGS. 1-6 show a card edge connector **100** of a preferred embodiment of this present invention, the card edge connector is mounted on a circuit board **200** and secured on the circuit board **200** by at least one bolt **30** (as shown in FIG. 5). The card edge connector **100** includes an insulating housing **10** extending along a longitudinal direction, a plurality of terminals retained in the insulating housing **10** and nuts **40**. The nuts **40** are fixed in the insulating housing **10** for bolts **30** to pass through, thereby bolting the card edge connector **100** on the circuit board **200**.

[0014] Referring to FIGS. 2-4, the bolts **30** pass through the circuit board **200** across through holes

201 defined on the circuit board and are screwed into the nuts **40**. To be detailed, the bolts **30** pass through the circuit board **200** upwards from a lower face of the circuit boards and finally fixed to the nuts **40**. Referring to FIGS. **3-4**, the insulating housing **10** includes two side walls **11** and a card slot **12** opening upwardly between the two side walls **11**. A couple of support portions **13** extends outside in a lateral direction perpendicular to the longitudinal direction from the side walls **11**, the support portions are located at opposite longitudinal ends of the side walls **11**. Each support portion **13** includes a notch **14** opening upwardly for receiving the nut **40** and a passageway **15** extending downwardly from the notch **14**. The shape of the notch **14** is fit to the nut **40**, so that the nut is un-rotated in the notch **14**. The bolt **30** goes through the through hole **201** of the circuit board **200** and the passageway **15** of the support portion **13** until the bolt **30** is screwed into the nut **40**. As a result, the insulating housing **10** is secured on the circuit board **200** stably.

[0015] Referred to FIGS. **4-5**, the side wall **11** includes a sliding groove **111** recessed from an outer face thereof, opening upwardly and communicating downwardly with the notch **14**. The notch **14** is of a hexagon with six sides. The shape of the sliding groove **111** matches a part shape of the notch **14** when viewed in an upper to lower direction, so that the nut **40** is installed into the notch **14** along the sliding groove **111** downwards. The card edge connector **100** includes a metal shell **50** covering outside. The metal shell **50** includes a horizontal portion **51** attached to the top surface of each support portion **13** and a vertical portion **52** vertically bending downward from the horizontal portion **51**. The vertical portion **52** has a board pin **53** extending downwardly. Therefore, the horizontal portion **51** helps to secure the metal shell **50** on the insulating housing **10**. What's more, the board pin **53** strengthens the stability of the insulating housing **10** and secures the insulating housing **10** on the circuit board **200**. In addition, as shown in FIGS. **3-5**, the notch **14** opens upwardly and the top surface **401** of the nut **40** is exposed to an exterior. The horizontal portion **51** covers the top surface of the nut **40**. Therefore, the nut **40** is limited from moving upwards. The notch **14** opens outside laterally, the vertical portion covers one of the six sides to limit the nut in the lateral direction. The metal shell **50** covers the insulating body **10**, and is welded with the top surfaces **401** of the nuts **40** by spot welding. Specifically, the horizontal portions **51** are spot welded to the top surfaces **401** of the nuts **40** and each forms a solder joint **54**. In alternative embodiment, a laser welding is used to replace the spot welding. In addition, an opening **511** is arranged at the junction of the horizontal portion **51** and the vertical portion **52**, and the opening **511** exposes the nut **40** and communicates with the exterior on the top surface of the nut **40**. The horizontal portions **51** are welded with the top surfaces **401** of the nuts **40** and the board pins **53** further extend downwardly from the horizontal portions **51** so that the nuts **40** and the bolts **30** are fixed on the insulating body **10**, and the whole card edge connector **100** is stabilized on the circuit board **200**.

[0016] As shown in FIG. **4** and FIG. **6**, one end of the insulating housing **10** is pivoted with a locking member **60**, the support portion **13** is located adjacent to the locking member **60**. The bottom surface of the supporting portion **13** has a raised portion **16** protruding towards the circuit board **200**. The passageway **15** penetrates the raised portion **16** downwardly. The raised portion **16** contributes to support the insulating housing **10**.

[0017] The above-mentioned embodiments are only preferred embodiments of the present invention, and should not limit the scope of the present invention, any simple equivalent changes and modifications made according to the claims of the present invention and the contents of the description should still belong to the present invention.

Claims

1. A card edge connector assembly comprising: a circuit board having two holes; two bolts; a longitudinal insulating housing including two side walls extending along a longitudinal direction and two support portions at a first longitudinal end of the side walls, each of the support portions having a notch opening upwards and a passageway extending from the notch to a bottom of the

support portion; a plurality of terminals retained in the insulating housing; and two nuts received in corresponding notches; wherein the bolts extend from a bottom of the circuit board across corresponding holes and corresponding passageways and are retained with corresponding nuts to secure the insulating housing to the circuit board.

2. The card edge connector assembly as claimed in claim 1, wherein the side wall includes a sliding groove recessed from an outer face thereof, and the sliding groove opens upwardly and communicates downward with the notch, and the sliding groove and a part of the notch are aligned as viewed in an upper-to-lower direction.

3. The card edge connector assembly as claimed in claim 1, further comprising a locking member located at the first longitudinal end of the insulating housing.

4. The card edge connector assembly as claimed in claim 3, wherein a raised portion protrudes from the bottom of the support portion towards the circuit board.

5. The card edge connector assembly as claimed in claim 3, wherein the notch is of a hexagon shape, and the nut has a matched shape with the notch.

6. The card edge connector assembly as claimed in claim 3, further comprising a metal shell, wherein the metal shell comprises two opposite side plates attached the side walls, a horizontal portion bending from each of the side plate and a vertical portion bending from the horizontal portion, and a board pin extending from the vertical portion, the horizontal portion covers a top surface of the support portion, and the vertical portion covers a side face of the support portion.

7. The card edge connector assembly as claimed in claim 4, wherein the notches open upwards and outwards, the horizontal portion covers a top of the nut, and the vertical portion covers a side of the nut.

8. A card edge connector adapted for being mounted on a circuit board, comprising: a longitudinal housing having a card slot extending along a longitudinal direction thereof; a plurality of terminals retained in the insulating housing; a locking member located on a first longitudinal end of the housing; two nuts; and two support portions integrally extending outwards in a lateral direction perpendicular to the longitudinal direction from opposite sides of the housing and located proximate to the locking member; wherein each of the supporting portion has a notch and a passageway opening from the notch and running through the support portion, and the nuts are received in the notches respectively.

9. The card edge connector as claimed in claim 8, further comprising a metal shell surrounding the housing and covering the support portion, and wherein the nut is sandwiched between the support portion and the metal shell.

10. The card edge connector as claimed in claim 9, wherein the metal shell and the nut are spot-welded.

11. The card edge connector as claimed in claim 9, wherein the metal shell and the nut are secured by laser welding.

12. The card edge connector as claimed in claim 8, further comprising a metal shell, wherein the metal shell comprises two side plates attached to the housing in the longitudinal direction, a horizontal portion bending from each of the side plates, a vertical portion bending from the horizontal portion, and a board pin extending from the vertical portion, the horizontal portion covers a top surface of the support portion, and the vertical portion covers a side of the supporting portion.

13. The card edge connector as claimed in claim 12, wherein the notches open upwards and outwards, the horizontal portion covers a top of the nut, and the vertical portion covers a side of the nut.

14. A card edge connector adapted for being mounted on a circuit board, comprising: an insulating housing extending along a longitudinal direction thereof; a locking member pivotally connected to a first longitudinal end of the insulating housing in the longitudinal direction; a plurality of terminals retained in the insulating housing; a metal shell covering the insulating housing; and two

support portions extending from the insulating housing and located proximate to the locking member; wherein the metal shell has two portions attached to top surfaces of the support portions and extending to the circuit board.

15. The card edge connector as claimed in claim 14, wherein a raised portion protrudes from each of the support portions towards the circuit board.

16. The card edge connector as claimed in claim 14, wherein the metal shell comprises two side plates attached to the housing in the longitudinal direction, a horizontal portion bending from each of the side plates, a vertical portion bending from the horizontal portion, and a board pin extending from the vertical portion, the horizontal portion covers a top surface of the support portion, and the vertical portion covers a side of the supporting portion.

17. The card edge connector as claimed in claim 16, wherein each of the supporting portions has a notch opening upwards for receiving a nut fixed to a bolt to secure the card edge connector to the circuit board.

18. The card edge connector as claimed in claim 17, wherein the notches open upwards and outwards, the horizontal portion covers a top of the nut, and the vertical portion covers a side of the nut.

19. The card edge connector as claimed in claim 18, wherein the horizontal portions are welded to the top surfaces of the nuts.

20. The card edge connector as claimed in claim 19, wherein an opening is arranged at the junction of the horizontal portion and the vertical portion, and the opening exposes the nut and communicates with exterior on the top surface of the nut.
